

More Solved Problems

1. Simplify the following equation by K-Map.

$$F(A,B,C) = \sum m(0,2,4) + \sum d(1,3,5,6,7)$$

BC	00	01	11	10
A				
0	1	X	X	1
1	1	X	X	X

$$F(A,B,C) = 1$$

2. Simplify $A(0,2,3,6)$ by K-Map

BC	00	01	11	10
A				
0	0		0	0
1				0

$$F' = BC' + AB' + A'C'$$

$$F = (F')' = (BC' + AB' + A'C')'$$

$$= (BC')' \cdot (AB')' \cdot (A'C')' \quad (\text{Theorem 5(a) extended})$$

$$= (B' + C') \cdot (A' + B') \cdot (A' + C') \quad (\text{Theorem 5(b)})$$

$$= (B' + C) \cdot (A + B') \cdot (A + C) \quad (\text{Theorem 3})$$

3. Simplify the following equation by K-Map

$$F(A,B,C,D) = \sum m(2,4,5,13,14) + \sum d(0,1,8,10)$$

CD	00	01	11	10
AB				
00	X	X		1
01	1	1		
11		1		1
10	X			X

$$F = AC' + BC'D + ACD' + B'C'D'$$

3.9 The Tabulation Method

The map method of simplification is convenient as long as the number of variables does not exceed five or six. As the number of variables increases, the excessive number of squares prevents a reasonable selection of adjacent squares.